

Python Regular Expressions (Regex) - Detailed Notes

Regex is a sequence of characters that defines a search pattern. It is used for searching, matching, validation, extraction, replacement and splitting text.

Import

```
import re
```

re.match()

```
import re
text='Python'
print(re.match('Py',text))
print(re.match('thon',text))
```

re.search()

```
import re
text='I love Python'
print(re.search('love',text))
```

re.findall()

```
import re
text='abc123xyz456'
print(re.findall(r'\d+',text))
```

re.finditer()

```
import re
text='abc123xyz456'
for i in re.finditer(r'\d+',text):
    print(i.group(),i.start(),i.end())
```

re.sub()

```
import re
text='abc123xyz456'
print(re.sub(r'\d+', '*',text))
```

re.split()

```
import re
text='apple,banana;orange grape'
print(re.split(r'[; ]',text))
```

re.fullmatch()

```
import re
print(re.fullmatch(r'\d+', '12345'))
```

Match methods

```
m=re.search(r'\d+', 'abc123xyz')
print(m.group())
print(m.start())
print(m.end())
print(m.span())
```

Metacharacters

```
. any char
^ start
$ end
* zero or more
+ one or more
? zero or one
{n},{m,n} repetitions
```

Character classes

```
[A-Z] uppercase
[a-z] lowercase
[0-9] digits
[^abc] not a,b,c
```

Shortcuts

```
\d digit\n\D non-digit\nw word\nW non-word\ns whitespace\nS non-whitespace
```

Groups

```
m=re.search(r'(\w+)\s(\d+)', 'John 25')
print(m.group(1))
print(m.group(2))
```

Alternation

```
print(re.findall('cat|dog', 'cat dog rat'))
```

Validation Patterns

```
Phone: \d{10}\nEmail: ^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$\nPassword: ^(?=[A-Z])(?=[a-z])(?=[0-9])(?=[\W_])[A-Za-z0-9\W_]{8,}$
```

Cheat Sheet

```
. ^ $ * + ? [] () | \d \D \w \W \s \S
```